

APPENDIX

A.0 Text and Preprocessing

This study analyses the text of Cicero’s letters *ad Atticum*. This selection of Cicero’s letters was chosen for two reasons: first, because the salience of the topic of production mode in this corpus allows for sufficient metadata to be generated from the text, and secondly, to guard against possible confounds related to the recipient of the letters. The text was downloaded from the Perseus Digital Library / Open Greek and Latin, displayed in the Scaife Viewer.¹ The text was downloaded as a TEI-XML file, and separated into per-letter rows in a CSV. Greek was kept in the text for the initial parse, and was taken into account when correcting the sentences manually. Epistolary formulae at the start of each text were also removed.

The unit for analysis was the individual letter. 67 letters could be labelled on the basis of internal evidence and Cicero’s metalinguistic comments, as being either written in Cicero’s own hand or dictated. Letters with fewer than 5 sentences were removed, leaving 65 letters overall: 46 AUTO and 19 DICT. The criteria for labelling were based on a close reading of the letters; ‘explicit’ and ‘inferred’ letters were not treated separately for the purpose of the analysis, but a small check for robustness was performed with only the explicitly labelled letters, which showed the same separation.

The syntactic features were analysed using the Stanza Universal Dependencies parser,² operating on raw text. The ‘main predicate’ of the sentence was taken as the parser’s ‘syntactic root’. For each sentence, the root position was taken as the 1-based index of the syntactic root (not including non-punctuation tokens), divided by the number of non-punctuation tokens in the sentence. Each sentence was then assigned a score between 0 and 1, with lower values indicating an earlier relative placement of the main predicate. For each letter, two measures of the variability of the main predicate placement across sentences were calculated: the proportion of sentences in the letter where the main predicate fell in the last third of the letter, controlled for letter length (LP), and the standard deviation of the position of the main predicate, again controlled for letter length (SD).

¹ Marcus Tullius Cicero. *Epistulae ad Atticum*. CTS URN: urn:cts:latinLit:phi0474.phi057.perseus-lat1. Accessed 25 January 2026 (Scaife Viewer release: v2025-11-07-001).

² Version 1.11.0: Qi et al. 2020.

For all the labelled letters, the ‘syntactic root’ was manually checked and corrected.³ The root was identified as either the main verb of the sentence, or, where no main verb was present (or the main verb was the copular) *esse*, the main predicate was identified. The annotations were performed by the author, blind to letter labels. Correction took place in 24% of cases (including misidentified sentence roots, and incorrect splitting of sentences⁴). The errors in the parser showed a systematic bias towards picking sentence roots early in the sentence; this was particularly visible in AUTO letters.⁵ Following manual correction of the roots in the labelled groups AUTO and DICT, there was no significant difference in the overall separation between groups. In order to make sure that labelled letters could be compared with unlabelled letters, the uncorrected parser scores were used. Given that AUTO-letters were more likely to have predicates identified too early, the parser errors work against the observed separation.

A.1 Enrichment with ‘correction’

The following table shows the effect of testing the features on the enriched ‘tails’ once a correction factor is applied from the estimate of the proportion of dictated letters in REST, that is $\pi = 0.30$.

Table 4: full ‘tails’ test (corrected, $\pi = 30$). p-values given for each feature are given below.

Feature	5%	10%	15%	20%	25%
λ	0.907	0.840	0.771	0.709	0.652
n per tail	24	49	74	99	124
<i>velim</i>	0.002*	0.0002*	0.0001*	<0.0001*	0.0002*
exclamations	0.112	0.015*	0.060	0.190	0.093
<i>nec_ratio</i>	0.069	0.044*	0.115	0.224	0.230
<i>ac_ratio</i>	0.220	0.004*	0.001*	0.030*	0.313
<i>si_et</i>	x ⁶	0.340	0.003*	0.001*	0.0004*

³ Checks were made on 1976 sentences.

⁴ For instance, on some occasions, the parser treated the ‘.’ in ‘Cn.’ as the full stop at the end of a sentence.

⁵ When corrections were made, the root was moved later 68% of the time for AUTO letters (219/324), compared with 56% (73/131) in DICT letters.

⁶ Wrong direction.

fut imperative	x	0.487	0.563	0.283	0.150
long perfect	1.000	1.000	0.497	x	x
<i>ego autem</i>	0.234	0.204	0.166	0.537	0.410
<i>qq_adv</i>	x	0.715	0.027*	0.048*	0.184
Direction	5/9	9/9	9/9	8/9	8/9

A.2 Multiple feature-testing correction

The exploratory phase on the labelled data was intentionally wide-ranging, and the features retained for further testing were motivated by an evaluation of apparent significance and linguistic motivations, taking into account possible confounding factors. Since these results were subject to confirmation tests on the enriched data, it was not necessary to account for the fact that numerous features had been tested at the discovery stage. However, the retained candidate-features were tested as a group, and so it was necessary to account for the likely ‘false-discovery’ rate when assessing statistical significance. The standard procedure for this is to use a Benjamini-Hochberg correction ($q = 0.05$) on the p-values for continuous features (and Fisher p for binary). Features where the direction was incorrect on the initial enriched test at the 10% threshold are included as part of the correction (so $n = 11$) and evaluated as $p = 1.0$. The following table shows the BH-corrected significant individual features at different thresholds in the ‘doughnut’ test (the independent confirmation), with the assumption of $\pi = 0.30$ and appropriate correction.

Table 5: BH correction for the ‘doughnut’ test

Individually significant features significant following BH correction (‘doughnut’ test)	
Threshold	Features
10%	1: <i>ac_ratio</i>
15%	2: <i>ac_ratio</i> , <i>si_et</i>
20%	2: <i>si_et</i> , <i>velim</i>

The number of individually significant features, after this multiple-feature correction, is low, but is not the focus of the argument in Section 3.4, which is principally concerned with the collective strength of the linguistic profile. To evaluate this strength as a whole, LP scores for REST letters were randomly permuted (with 2,000 iterations); the ‘doughnut’ tails were reconstituted based on these permutations, and the directions of each feature were then

recomputed. The following test asks how often a random sorting of the REST letters produces correct direction.

A.3 Permutation tests for the ‘doughnut’

The Benjamini-Hochberg correction applied above identifies 1–2 individually significant features at $q = 0.05$, depending on threshold. However, the argument of §3.4 rests primarily on the collective directional consistency of the feature bundle, rather than on any single feature reaching significance after correction. BH assumes independence among features under the null; since the features are expected to be positively correlated under the alternative hypothesis (all being driven by the same latent variable), BH is conservative for this setting. A permutation-based test provides a more appropriate framework for evaluating the bundle as a whole. The results are given in Table 6.

The null hypothesis is that the syntactic sorting variable (LP) is unrelated to the lexical and morphological features.

Table 6: ‘doughnut’ permutation test: p = the proportion of 2,000 permutations which give at least the observed direction count. The null hypothesis is that LP (the syntactic sorting variable) is unrelated to the profile of lexical and morphological features.

Threshold	Observed	Null mean	p
10%	8/9	4.0	0.011
15%	9/9	4.2	0.001
20%	7/9	4.2	0.074
25%	9/9	4.2	0.0025

The 15% threshold is particularly notable here: 9/9 features move in the expected direction with $p = 0.001$. Two features flip at the 20% threshold, but return to the direction predicted at 25%. However, with the focus still on the power of the profile over individual features, a permutation test that accounts for the magnitude of each effect within the profile as well as its direction gives stronger results across both ‘doughnut’ tails. The following test sums the z-scores of each feature across both tails (positive whenever the feature is in the direction as predicted; negative otherwise). The results are shown in Table 7. This score remains significant across all four thresholds. The features, as a collective profile, discriminate effectively between the two groups even where individual directions of binary features are affected by noise in the data.

Table 7: ‘doughnut’ permutation test (signed z-score sum). p = the proportion of 2,000 permutations which give at least the observed sum of the z-scores of the observed features (Σz). The null hypothesis, again, is that LP is unrelated to the profile of lexical and morphological features.

Threshold	Observed Σz	Null mean	Null SD	p
10%	12.80	+0.01	3.92	0.001
15%	13.38	+0.02	3.84	<0.001
20%	10.97	−0.03	3.85	0.003
25%	12.59	−0.06	3.97	<0.001

A.4 ‘Tails’ (10%) within chronological windows

‘Predicted direction count’, and AUC based on composite z-score for all features. ‘SB’ refers to the Shackleton Bailey (chronological) numbering for each letter; ‘SB gap’ = aggregate SB number for DICT-like / AUTO-like letters. The results show that there is a statistically significant separation between the two groups in every chronological window tested.

Table 8: ‘tails’ test within chronological windows

Window	nA	nD	SB gap	Dir.	Binom p	AUC	Perm. p
All tails	31	31	+54	8/8	0.004	0.736	<0.001
SB 50–150	16	7	+6	6/8	0.145	0.674	0.103
SB 100–300	15	19	+17	7/8	0.035	0.763	0.004
SB 150–300	10	15	+9	6/8	0.145	0.740	0.020
SB 200–350	10	16	+3	6/8	0.145	0.791	0.008
SB 250–400	8	13	−6	8/8	0.004	0.856	0.002
SB 300–430	6	9	−6	7/8	0.035	0.778	0.038